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# Queen's University Belfast

School of Electronics, Electrical Engineering and Computer Science

## ECS8056

Themed Research Project

Delivery 2024-2025

E-MAIL  
[l.bai@qub.ac.uk](mailto:l.bai@qub.ac.uk)

WEB  
[www.qub.ac.uk/eeecs](http://www.qub.ac.uk/eeecs)



# Introduction

## Overview

The Themed Research Project represents a major piece of work on the pathway to a MSc AI degree, and this is reflected in both the weighting in CATS points and the amount of support that is available. The specification for the Themed Research Project module indicates that it should represent enough work to occupy a student full-time for at least 12 weeks which equates to at least 24 weeks part-time.

For many students on the MSc pathway, writing a dissertation is a challenging experience. It is, almost by definition, a self-directed process. In the other modules of the course, you have been set targets and objectives through the coursework assignments and deadlines but in this phase of the course; you will need to set your own deadlines and manage the experience. You will need perseverance, endurance, and determination. Initially this can seem intimidating, but it is also an opportunity for you to learn and grow. Finishing the Themed Research Project on time is largely about having the will to do it. Most of the skills that you will need, you already have gained. However, the Themed Research Project is designed to be a learning experience and it is highly likely that you will extend your existing AI skills or that you will have to learn some new skills. Writing your dissertation will enable you to develop valuable technical writing skills. Regardless of the career path you have chosen, the processes of thinking analytically, assembling a well-reasoned AI solution, writing professionally, and managing your time will all serve you well in the future.

## Aim of ECS8056

The aim of the module is to consolidate the knowledge and skills acquired in the modules completed earlier in the programme and to address a specific research problem relevant to the field of AI, possibly involving collaborations with other disciplines such as health, biomedical sciences, environmental sciences, or social sciences. The project will focus on a topic derived from the thematic areas of AI, including Machine Learning (ML), Knowledge Engineering (KE), Computer Vision (CV), Natural Language Processing (NLP), and AI for Health (AI4H), or other themes approved by the Programme Committee. This will ensure students have the opportunity to build on prior work from their previous modules.

## Learning Outcomes

The general learning outcomes which you should achieve for this module are that on completion you should be able to:

1. Employ a range of communication skills, including the effective use of technology, to manage and disseminate knowledge to appropriate audiences.
2. Plan and implement change through research and project management, with adherence to ethical guidelines.
3. Design, implement and test a chosen solution according to professional best practice.
4. Analyse and critically evaluate concepts, arguments, and evidence in the literature.
5. Beware the current AI issues (e.g., trust, ethics, interpretability, human centred AI, and AI for good).

## What you have to submit

On completion of the module, you are required to submit two parts of the project submission.

1. A Research Paper (70%) must be submitted conforming to a stated specification. The research paper should be of approximately 5000 words, between 6 and 8 pages in length (strict limit of 8



pages), and in a format suitable for a conference submission. Detailed information on layout and format will be made available to the students at the beginning of the project. Some students may be concerned that their work is going to be too short or too long, but you can seek clarification from your academic supervisor.

2. Supporting materials (30%) must be submitted, which will consist of a written report, code repository on QUB EECS GitLab and, visual material. The visual material will aid reviewers with the assessment and include a recorded video demo.
3. Project Demonstration (0%), at the end of the project, each student is required to demonstrate their completed project to the supervisor and assessor in the same session. The demo and presentation should last no longer than 20 minutes, which will be followed by a question-and-answer session. The student should demonstrate the project to the supervisor and assessor and answer further questions to the extent to which the completed research work meets the original research objectives.

## Module organisation, timetabling and submissions

### Levelling up activities

Visit the ECS8056 Canvas website for up-to-date information on levelling up sessions.

### Submissions

Visit the ECS8056 Canvas website for up-to-date information on submission deadlines and other important dates.

## Project Allocation Process

The project allocation for ECS8056 takes place in Semester 2. A list of the proposed projects will be provided via Google Sheets. Students can look for interesting projects and contact the supervisors directly to discuss the project and the requirements (background knowledge and experience, what is expected in terms of AI techniques or programming, supervision style, etc.). Google Sheets allow sorting based on themes and topics, but these are meant mostly for guidance, so do approach a supervisor if you are interested in a topic even if it is not explicitly listed. Keep in mind that some supervisors get many requests from students and that there is a limit per supervisor. Once both the supervisor and the student agree, the supervisor should email the project coordinator (and CC the student) to register this choice. The choices must be provided by the given deadline to facilitate the project allocation. Without receiving your preferences by the deadline, you will be assigned a project from the (remaining) available projects. The final allocation is done by the project coordinator. The allocation is not finalised until the project coordinator confirms the allocation. While we will do our best to accommodate each student's choice, we ask your understanding in the limited resources and supervisors available in the School.

The students can also choose to propose their own project topic. The initial idea often needs to be refined together with a potential supervisor to make sure that it is relevant, doable, and challenging. Make sure to leave some extra time if you decide to go down this route – don't leave it to the last moment! In this case, the student writes 1-2 pages (500-1500 words long) summary of what they propose to do, how they propose to do it, and what the outcome will be. Then contact a potential supervisor (typically someone with expertise in the topic) asking them to supervise. If you are not sure who would make the best supervisor for a particular topic, contact the project coordinator to make some suggestions. Once the project is agreed, the supervisor should email the project coordinator (and CC the student).



## Assessment

The assessment for this module has several elements:

|                                             |             |
|---------------------------------------------|-------------|
| <b>Problem Specification / Project Plan</b> | 0%          |
| <b>Research paper</b>                       | 70%         |
| <b>Supporting material</b>                  | 30%         |
| <b>Project Demonstration</b>                | 0%          |
|                                             | <b>100%</b> |

In order to pass the module, you are required to attain an overall score of 50%. The completed project, with all its files (documents and recorded demo video), must be submitted via Canvas by the specified date.

These submissions are used for the assessment. **NOTE:** If the code and recorded demo video are not received then the project will not be considered by the Board of Examiners as completed.

Each student's work is assessed by two academics. One of these will be the academic supervisor, and the other is usually someone who has not had direct contact with the student during this module. The supervisor and the assessor will assign an agreed mark which will be returned to the student as the final mark. If there is a significant degree of disagreement (10%) between the two academics on the student's grade, a third opinion will be sought. The external examiner will moderate the marking of the projects.

The grade will correspond to:

- greater than 70% (distinction)
- 60 to 69% (commendation)
- 50 to 59% (pass)
- below 50% (fail)

After the meeting of the Internal Examiners and following the meeting of the Board of Examiners with the External Examiner, your grade will be published on QGIS.

The pass mark for this project is 50%. Students who fail the project will have the opportunity to resubmit capped at 50%. Students who pass the module may then register for graduation by clicking the 'Register' button on their QGIS homepage. The closing date for registration is usually posted on QGIS. Students who intend to graduate in absentia are also required to register and you should be aware that there is a fee for graduation.

### The Assessment Criteria

Because the nature of the work is so individual, it is not possible to provide a definitive list of criteria for a high-scoring Themed Research Project. However, the following guidelines will be used by your academic supervisor and industrial advisor (if any) to grade your work:

#### Problem Specification / Project Plan

The student has to demonstrate that they understood the problem and described a sensible approach to solving it and that they clearly specified the objectives of the project. There should be an analysis of the problem domain and content of the project. You could, for example, the background and context, outline the research methods and AI techniques that will be employed to address the problem and state the anticipated results or contributions of the research. A timetable for the work should be included along with



notes on potential problem areas. The primary purpose of this deliverable is to ensure that the project is suitable/feasible, and that the student has a clear plan and is engaging correctly with the project. This document should be no more than five pages in length. A copy of the Problem Specification/Project Plan should be uploaded to Canvas ECS8056 by the deadline for submission as outlined on ECS8056 Canvas website.

#### Research paper (worth 70%)

The main priority for the research paper lies in providing valuable information and outline the reasoning behind framework, architecture, and model decisions.

#### **Writing Style**

Use the word template provided, approximately 8 pages in length. Overall, the main text of the research paper should be clear, concise, fluent and well organised. You should concentrate on the quality of the research paper rather than the quantity. Spelling should be correct, sentences grammatical, and formulae, figures, and tables accurate. All figures, tables, and appendices should be given numbers and headings. As is usual in academic / technical writing, do not describe things in the first person (say what was done, rather than you did it).

Use of either Latex or Word is acceptable, however, only a Word Template is provided. All submissions, regardless of whether they are prepared with Latex or with Word, should be in PDF format.

#### **Title Page**

The title page should have the following format:

‘Title of Dissertation’  
A dissertation submitted in partial fulfilment of  
The requirement for the degree of  
MASTER OF SCIENCE in Artificial Intelligence  
in  
The Queen’s University of Belfast  
By  
‘Student Name’  
‘Date of Submission’

#### **Declaration Cover Sheet (Declaration of Academic Integrity)**

This is available at the back of this document; it must be signed and submitted with the document.

#### **Acknowledgements**

This is a section written to those who have helped you, as the author, during the preparation to completion of the project and dissertation. The student should also thank anyone who has given you financial support.

#### **Abstract**

This is a summary (200 words) outlining a brief summary of the problem to be studied, the objectives of the study, the proposed methodology, and the expected results or achievements.

#### **Keywords**

List up to a maximum of 5 keywords that captures the research.



### ***Main Body of the Dissertation***

The following section outlines the sequence of chapters that should be included in the dissertation. Some dissertations may be organised differently from the section headings given below. However, you should attempt to address each of the following aspects in your dissertation. Further advice will be available from your supervisor.

#### ***Section 1: Introduction***

The student must clearly describe the perceived problem and introduce the area of research, the problem being investigated including how its positions within the area of research, and the aim of your research. Clearly state the objectives of your MSc AI project. What do you aim to achieve or contribute with your research. Define the scope of your project by specifying the boundaries and limitations, ensuring that the reader understands what aspects will be covered. Include specific research questions or hypotheses that your project seeks to answer or test. These questions should align with your objectives and guide the reader's expectations. Provide a brief overview of the methodology or approach you have taken to address the research questions. Briefly outline how the rest of the paper is organised. Mention the main sections, such as literature review, methodology, experiments, results, and conclusion.

#### ***Section 2: Literature review***

This section should be no more than 1 page of text include a systematic literature review of research articles. Review recent and state-of-the-art approaches, methodologies, and technologies relevant to your project. Discuss the strengths and limitations of these approaches. You need to make sure with your supervisor that the articles you are reviewing are worthy of referencing in an MSc project.

For each article considered in the review you may consider: (i) the approaches used by other researchers to solve the problem, (ii) the merit and weakness of the approaches, and (iii) any identified gaps, and (iv) how will your approach be different from existing approaches and how it will bridge any identified gap.

#### ***Section 3: Methods***

This section should detail the specific steps and procedures you employed to conduct your research. You may consider the following:

- (i) provide details about the dataset(s) used in your experiments. Include information such as the source, size, characteristics, and any preprocessing steps applied. If you created a custom dataset, explain the reasons behind its creation and any considerations taken during the process.
- (ii) Describe the architecture of the AI models you employed in your project. This includes the type of model (e.g., neural network, decision tree), the number and type of layers, and any unique characteristics. If your project involves multiple models or variations, provide details for each.
- (iii) Outline the training procedure for your models. Include details such as the optimisation algorithm, learning rate, batch size, and number of epochs. Discuss any regularisation techniques or strategies used to prevent overfitting.
- (iv) If you conducted hyperparameter tuning, describe the methods and criteria used for optimisation. Explain the rationale behind the chosen hyperparameters.
- (v) If your project involves sensitive data or raises ethical concerns, discuss the steps taken to address these considerations. Include information on obtaining consent, data anonymisation, or any ethical review processes.

#### ***Section 4: Experimentation and Results***

This section is where you present the details of how you conducted your experiments, the methodologies you employed, and the outcomes of your research. In this section, you may include the following:



- (i) Detail the hardware and software environment used for conducting your experiments, including the specifications of the computing resources and software libraries/frameworks. If there are any dependencies or specific configurations, provide this information for reproducibility.
- (ii) Clearly state the metrics used to evaluate the performance of your AI models. Explain why these metrics were chosen and how they align with your research objectives.
- (iii) If relevant, discuss any limitations or considerations associated with the chosen evaluation metrics.
- (iv) Present the results of your experiments in a clear and organised manner. Use tables, figures, and graphs to illustrate key findings. Include both quantitative and qualitative results, highlighting any patterns, trends, or noteworthy observations. If applicable, provide statistical analyses to support your findings.
- (v) Compare your results with baseline models or existing literature. Discuss how your approach performs in comparison and highlight any improvements or novel insights. If there are unexpected outcomes, address them and provide potential explanations.

### ***Section 5: Discussions***

This section should interpret the results in the context of your research questions and objectives. Discuss the implications of your findings and their significance. Address any limitations or challenges encountered during the experiments.

### ***Section 6: Conclusion and Future Work Considerations***

Conclusions should include a summary of the project work and findings arising from the investigation. Finish this section by highlighting the limitations of your work and making recommendations on how the work can be further extended. Discuss any unresolved questions or limitations that could be addressed in subsequent studies.

### ***References***

Up to a maximum of 35 references for the entire article.

References - is a complete list of the articles or other information sources that have been cited/used in this article, giving the publication details for each reference (e.g., title of the paper, name of the journal or conference, time/location it is published, volume/issue numbers, page numbers, name of the publisher, etc.).

Note: Wikipedia and similar are not acceptable sources to reference. Keep the use of web pages to a minimum, use only when necessary.

Recommended method for referencing is the Institute for Electrical and Electronics Engineers (IEEE) format - \IEEE citation style includes in-text citations, numbered in square brackets, which refer to the full citation listed in the reference list at the end of the paper. The reference list is organised numerically, not alphabetically.

Referencing examples:

[1] Y. Ren, C. Zhu, and S. Xiao, "Object Detection Based on Fast/Faster RCNN Employing Fully Convolutional Architectures," *Mathematical Problems in Engineering*, vol. 2018, pp. 1–7, 2018, doi: <https://doi.org/10.1155/2018/3598316>.

[2] G. Grassi and L. A. Grieco, "CNN-based architecture for real-time object-oriented video coding applications," *International Journal of Circuit Theory and Applications*, vol. 33, no. 1, pp. 53–64, Jan. 2005, doi: <https://doi.org/10.1002/cta.303>.



### **Appendices**

(It is optional and is included within the maximum page limit) – should include other information which is not convenient or is disturbing to be included in the main body of the research paper (e.g. a proof of a mathematical formula that is used in the research etc.).

### **Supporting Materials (worth 30%)**

It is recognised that an eight-page research paper may not be able to contain full details of the results achieved, methodology followed, or alternative solutions explored due to space limitations. Accordingly, the student must submit supporting material to assist the reviewer. The supporting material should cover the following:

- An in-depth investigation of the context and literature to complement that within the paper.
- A clear description of the stages of the project life cycle undertaken, including a description of how verification and validation are conducted.
- A description of tools used to support the development process of the project.
- Critical appraisal of the project, indicating the rationale for any design/implementation decisions, lessons learnt during the project, and evaluation (with hindsight) of the project outcome and the process of its production.

The supporting material will consist of a written report, code repository on QUB EECS GitLab and, visual material. The visual material will aid reviewers with the assessment and include a recorded video demo.

### **Guidance for documentation on the GitLab repository**

The academic supervisor should have access to the student's QUB EECS GitLab repository to ensure the amount of individual contribution from the student is acceptable. Access to a repository of code applies in all project cases whether individual, academic or company based. The details of the repository should be agreed at the outset of the project.

If a company project is undertaken and it contains confidential / sensitive information, then there must be an agreement that the academic supervisor and assessors can view and assess the project code. Therefore, if confidentiality / non-disclosure forms have to be signed they should be submitted to the supervisor at the start of the project.

The complete program code developed during the research project should be uploaded to QUB EECS GitLab: <https://gitlab.eecs.qub.ac.uk/>; the student must provide access to both the supervisor and the assessor. The program code must be appropriately annotated with comments and each component should have a complete module header. Programs should be written in a clear style with good program structure and well-defined data structures.

It is the student's responsibility to become familiar with using git. Guidance can be found at the self-service site (<https://selfservice.eecs.qub.ac.uk/docs/git/>)

### **Recorded Video Demo (Project cannot be marked without Video)**

The project should be demonstrated by recording screen captures of it in use with supporting appropriate audio commentary for viewing by the External Examiner. The recorded video demo should show how your project works with examples of input data, how it is trained, evaluated, used by users, what results it gives, and what future plans are. The video must **not exceed ten minutes** in runtime. At the start of the video, you should show the title of your project along with your name, student number and image of yourself.



The video should be saved in **MP4 format**. Students must ensure that the video plays on VLC on the lab computers in the Computer Science Building (CSB) labs. You should test your video on the CSB lab computers before uploading your video through Canvas.

All text on screen in the video must be clearly legible and audio commentary must be clearly audible.

The Video Demo will be made available to the External Examiner as well as been reviewed by the other assessors. It may be made available to future students of the course.

**Please Note:** If the recorded video demo is not submitted as part of the final submission or cannot be played in full by the assessors, then the project submission may not be marked.

#### Project Demonstration

The student should arrange in advance a suitable time and date with their supervisor to present/demonstrate their project in a professional manner after the deadline for the submission of the dissertation. This may include communication with companies, if present for employer-based projects, and other academic assessors. The student should bring all required software and hardware to the demonstration. As the demonstration may require setting up of hardware and/or connecting to servers, demonstration of the software and the answering of questions posed by assessors, the student should ensure that they provide themselves ample preparation time for their demonstration.

## Ethics

Here you should discuss any ethical considerations pertaining to your project. If you are planning to work with people (especially children), animals, sensitive private data, or if there are other considerations, you should discuss them here and explain how you went about obtaining necessary approval (any Ethics applications). Refer to University's policy on the Ethical Approval of Research:

<https://www.qub.ac.uk/Research/Governance-ethics-and-integrity/FileStore/Filetoupload,915687,en.pdf>

## Supervision

One of the key differences between the MSc AI Themed Research Project and all the other assignments you have completed as part of the MSc Artificial Intelligence is in the way you will be tutored. Each student on this module is assigned to a supervisor for the duration of the project. Your supervisor is responsible for providing advice and guidance to permit you to complete the project in the time available.

The Themed Research Project should be a period of self-directed study, so your academic supervisor cannot get in the way of this goal by becoming too involved. They are not expected to tell you what to do, chase you for work, structure your studying, or design your project. They will offer advice and support, check that you are on the right lines and point you in the direction of relevant materials for both your project and dissertation.

At the beginning of the project, you and your supervisor should agree on a project timetable and arrange a series of meetings where progress, problems and writing can be discussed. You should make sure that your supervisor is kept up to date with the progress of the project as well as the dissertation. Your supervisor will offer general guidance and support. You may submit a draft of your research paper for your supervisor to comment and provide feedback on the draft. Your supervisor may provide feedback on the content and structure of the draft however it is not their role to proofread the document for spelling and grammatical



errors. Upon receiving comments and feedback on a draft chapter, it is not acceptable to resubmit a revised draft for further feedback. It is important that if you wish to receive feedback on the draft work, that it is submitted to the supervisor in good time to allow this to happen.

## Themed Research Project Guide

### Responsibilities

You are expected to take responsibility for your own studies, and it is up to you to arrange access to whatever equipment and resources (e.g.) that you need.

It is your responsibility that:

- your problem specification and project plan are submitted on time
- your project is completed according to plan and on time
- problems are sorted out at an early stage
- your project is "debugged" and working as specified
- your finished project and code are delivered through Canvas on time with all associated files and is working correctly
- your research paper is submitted on time
- regular meetings are held with your academic supervisor (don't hide from them because you haven't done as much work as expected)
- advice and guidance are sought from your academic supervisor at an early stage
- all your written work is proofread and corrected before submission

It is in your own interest to maintain regular contact with your academic supervisor for the duration of the Project, and to have a good working relationship with them. It's certain that at some period during the Project your academic supervisor will be unavailable due to holidays or work commitments. You should take account of your academic supervisor's schedule in your planning and not expect them to be always available when you need advice and guidance. You should also give them ample time to reflect on draft work which you are submitting. It is wise to send regular updates and draft pages to your supervisor rather than sizable pieces of work.

Although most students quickly establish a good working relationship with their academic supervisor, there are occasionally cases where this does not happen. If you are concerned about the level of supervision you are receiving, you should in the first instance raise this with the Project Coordinator. If this action does not lead to a satisfactory conclusion, or you would rather not approach the Project Coordinator, then you should acquaint the Course Director for the MSc Artificial Intelligence with your concerns. Note that the School and University expect you to raise any difficulties which you have with your supervision in a timely manner.

### Minimum Contact

Normally a frequency of one meeting per two weeks should be regarded as an effective level of engagement with your supervisor/company mentor.



## Recording Meetings

It is a university Master's level requirement that you keep minutes of meetings with your academic supervisor. The minutes should be sent to your supervisor within 48 hours of the meeting and before the next meeting. The minutes should contain a brief summary of the meeting, the matters discussed, and any action agreed between you and the supervisor. At the next meeting, the record of the minutes from the previous meeting should be reviewed and discussed.

## If Things Go Wrong

Your academic supervisor also has a pastoral role. This is especially important if you are experiencing difficulties with the work or in your life. The default position of all the academic supervisors on the course is that we will assume things are going smoothly unless you tell us otherwise.

For some students, this phase of the course can be a lonely and frustrating time. Some students experience a crisis of confidence mid-way through the work. One way of overcoming this is to break the work down (for all the components - project and dissertation) into small steps and tackle them one at a time. Motivation may be maintained and improved as each step is "ticked off". Making consistent progress (even if it is only a small piece every day) is a better strategy than trying to "blitz" the work as you approach the deadline. If you are experiencing a crisis of confidence, talk things over with your academic supervisor. You may be surprised at how helpful such a strategy can be.

It is not uncommon for students to experience personal problems ranging from ill health to family bereavement. There are procedures available to help students overcome such difficulties. It is rare for the School to be confronted with something entirely new, and often we can provide tangible help very quickly. Advise your academic supervisor or speak to the Course Director if you are having personal problems which are impacting on your work. Some of these difficulties may be categorised as Exceptional Circumstances and how the School may be officially informed of these circumstances may be found at <https://www.qub.ac.uk/directorates/AcademicStudentAffairs/AcademicAffairs/GeneralRegulations/Procedures/ExceptionalCircumstancesProcedure/>

Some students may decide during their project that they do not wish to continue with the work. Any student on this phase of the course is, by definition, entitled to ask to transfer to the Postgraduate Diploma pathway, and to accept the Diploma. A request to transfer from the MSc to the Postgraduate Diploma pathway has to be in writing. Please do not take such a decision lightly. If you think that things are getting too much for you, make an appointment to meet the Course Director.

## Starting the work

Use the early days of the Artificial Intelligence Themed Research Project as a time for research and experimentation. See what is available in your project domain and what ideas you can borrow. Look for unusual and effective ways of solving your problem and see if you can incorporate some of these into your proposed method.

You should also use the early weeks to consider what content you will incorporate into your project and discuss your ideas with your supervisor.

## Avoiding Plagiarism

During the course of the MSc Artificial Intelligence, you will have explored some of the issues relating to academic plagiarism. Doing original work, especially on something as protracted as a project and



dissertation at Master's level is not easy. It is usual practice in academic writing to build upon the ideas of others. The use of another's work does not constitute plagiarism unless and until that work is presented as being your own. In the pressure to complete the dissertation on time there is always the temptation to take a short-cut without realising that these may lead to suggestions that the submitted work is not original. The satisfaction in completing the Project comes from mastering the complex material which underpins the field of Artificial Intelligence.

You must always distinguish your own ideas, words, comments, and contributions in the Project from those of others. To do so requires the high level of intellectual honesty which is expected from you and your peers.

When you restate another person's words, thoughts, or ideas you are paraphrasing them. In such a case you are still obliged to reference the original work. Paraphrased work does not have to be enclosed in quotation marks, but the source still needs to be fully referenced.

A summary is a looser form of paraphrasing, but this also should be referenced. A summary involves condensing and rephrasing other authors' work and is an important facet of academic writing. Using appropriate referencing should make clear to the reader which ideas come from which author.

### Originality of Computer Code

The Project is your chance to demonstrate what you have learnt from the taught modules, and to show your AI skills. As part of this process, you will inevitably have to spend some time either revising or learning new skills in coding.

You will only be given credit for code which you have written, and which is original to the project. One of the aims of the module is to determine students' level of skill and this would be compromised if "cutting and pasting" code from other developers was accepted. Code which you have developed for previous assignments which is recycled for the project will not attract marks.

Almost every project borrows ideas from elsewhere, and you are encouraged to reuse the exercises used in the practical classes as springboards for work on your project. However, you will not receive any credit for directly using exemplars and practical exercises from class, textbooks, web or other sources.

### Submitting Your Finished Work

As your work nears completion you should look at the dissertation "in the whole". Check that the separate sections link together and that any changes made in the methods and experimentation during the implementation of the project are reflected in the dissertation. Give your supervisor a chance to see and comment on your project as you are working on it. It is unwise to keep the project from your supervisor and then spring it on them as a fait accompli a few days before submission.

You should submit your project (research paper and supporting materials) via Canvas as part of the final submission.

### Project Schedule

The schedule for the project and milestones is available on Canvas.

## Regulations

The completed project and dissertation have to be submitted by the deadline for submission date. Standard university regulations on late submissions apply.



See, <https://www.qub.ac.uk/directorates/AcademicStudentAffairs/AcademicAffairs/GeneralRegulations/StudyRegulations/StudyRegulationsforPostgraduateTaughtProgrammes/>

The project and all ancillary files must be provided to the School via Canvas by the deadline for submission date. Both components (research paper and supporting materials) constitute the submission, late submissions apply to the whole submission.

A student may not be given credit for any work submitted in whole or in part for another assignment or part of an assignment within the modules associated with the MSc Artificial Intelligence. Materials submitted as part of the assessment for the MSc Artificial Intelligence may not be concurrently submitted as part of another degree at The Queen's University of Belfast or any other university.

A signed and dated declaration sheet asserting the originality of the work must accompany each copy of the dissertation and should be inserted after the title page of each copy of the dissertation submitted.

The written component of the Project should be submitted in English. The spelling should be correct, sentences grammatical and formulae, figures, and tables accurate.

All figures and tables included in the dissertation should be labelled and referenced from the main text.

Computer code should be presented in code repository on QUB EEECS GitLab and include comments. As far as possible all work for the project must be original. All non-original material must be acknowledged in the appendix. It is acceptable to use computer code from other people or which is in the public domain, provided that the source is made absolutely clear by comments within the code and noting the source in the appendix.

In some cases, and at the discretion of the Board of Examiners for the MSc Artificial Intelligence students may be required to attend a viva voce examination before their Project is accepted.

An instance where work from another student's submissions or unpublished work from this or other institutions has evidently been incorporated into the Project will be considered a serious breach of academic regulations.

If work submitted by a student is considered not to be original, and is not properly attributed to the original author, the examiner is obliged to bring the matter to the attention of the Chair of the Board of Examiners. If the Chair of the Board of Examiners is satisfied that unattributed and non-original work has been presented as that of the student, the work will be brought to the attention of the School Director.

The Director will cause an investigation to be brought into the work. At this stage, the student will be invited to explain the submission of apparently non-original work. If the Investigation Panel is not satisfied by the student's explanation, then the work will be brought to the School Board of Examiners with a recommendation that:

- the submission should not be accepted (in which case the student will be shown as failing the Project and Dissertation through non-submission), or
- a mark of zero is recorded for the work (in which case the student fails the Project), or
- the grade or mark should be substantially reduced.
- disciplinary action may be taken by the Board of Examiners.

The publication of any results of the student's work is left to the discretion of the supervisor. Material supplied at the outset of the project may be copyrighted solely for educational use.

## Scope and Level of Difficulty



The project is meant to bring out the best in you, and to stretch your ability. On the one hand, if you are given a problem which is beyond you, you may not be able to deliver anything much at all; on the other hand, if you are given a problem which is not the most challenging, it will not give you scope to do yourself justice, or the chance to gain very high marks. We want you to realise your maximum potential. Student-specified projects must also meet these constraints, by agreement with the supervisor.

While we will take your stated preferences into account in allocating projects, we cannot guarantee a project which meets all your preferences. Part of the training in this module is being able to take any task which is given to you and complete it to the best of your ability.

**SCHOOL OF ELECTRONICS, ELECTRICAL ENGINEERING and COMPUTER SCIENCE****ECS8056 – THEMED RESEARCH PROJECT**

A signed and completed cover sheet must accompany the submission of the Themed Research Project dissertation submitted for assessment.

Work submitted without a cover sheet will **NOT** be marked.

**Declaration of Academic Integrity**

Before signing the declaration below please check that the submission:

1. Has a full bibliography attached laid out according to the guidelines specified in the Student Project Handbook
2. Contains full acknowledgement of all secondary sources used (paper-based and electronic)
3. Does not exceed the specified page limit
4. Is clearly presented and proof-read
5. Is submitted on, or before, the specified or agreed due date. Late submissions will only be accepted in exceptional circumstances or where a deferment has been granted in advance.
6. Software and files are submitted via Canvas.

**I certify that that the submission is my own work, all sources are correctly attributed, and the contribution of any AI technologies is fully acknowledged.**

**I declare that I have read both the University and the School of Electronics, Electrical Engineering and Computer Science guidelines on plagiarism -**

**<https://www.qub.ac.uk/directorates/sgc/learning/LearningResources/Plagiarism/> - and that the attached submission is my own original work. No part of it has been submitted for any other assignment and I have acknowledged in my notes and bibliography all written and electronic sources used.**

**I am aware of the disciplinary consequences of failing to abide and follow the School and Queen's University Regulations on Plagiarism.**

**Name: (BLOCK CAPITALS)**

**Student Number:**

*Student's signature*

*Date of submission*

